



System Documentation

Diesel Engine

Engine Control System ECS

- Engine Control Unit ECU9

- Engine Ident Label EIL

12V/16V/20V4000Gx4

12V/16V/20V4000Gx3

E532474/01E

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1 Safety

1.1 Important provisions for electronic products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Assembly, maintenance, and repair without ESD („Electro Static Discharge“) equipment and ESD work station
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

1.2 Intended use of electronic products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original configuration of the delivery or in a configuration approved in writing by the manufacturer
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Intended use of gendrive engines

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and approval has been granted.

The section below applies to applications deployed in regions under the jurisdiction of the EPA.

- Use of EPA Tier 2 and Tier 3 certified genset engines.
- EPA Tier 2 and Tier 3 certified genset engines may only be used in generator sets for stationary emergency applications. Non-emergency operation is limited to 100 hrs per year.

The engines may be used in fixed speed engine-generator sets only. Permanent limitation to a max. engine power of less than 560 kW (e.g. by means of correspondingly low generator power) is not admissible.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.4 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:

- The installation of the engine
- The design of the engine-generator set
- The installation of the engine-generator set

The relevant standards are, in particular:

- DIN EN 1679-1
- EN ISO 8528-13

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. We therefore recommend a redundant design of the engine-generator set system in safety-critical applications.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, exhaust turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).

- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:

- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary and suitable safety devices to reliably exclude the possibility of personal injury, e.g.:

- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these protective devices. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the protective devices.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Chemical resistance

Only use hoses, pipes, fittings, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.

Incorrect synchronization

Load application with incorrect synchronization (generator connection with phase offset) can cause considerably high torque pulses. The plant integrator must design the control system such that the possibility of a load application with incorrect synchronization is excluded.

Emergency stop

The diesel engine has inputs for Stop and Emergency stop. The emergency-stop function must be executable in the whole plant independently of the operation stop. This is the only way to ensure that a safe shutdown of fuel injection takes place on a diverse redundant path and thus results in engine standstill. Through an appropriate design of the engine-generator set or whole plant, the operator of the engine must guarantee that safe, non-reversible stopping of the engine is as fast as possible, including all of its dangerous system components, when the emergency-stop actuating element is activated. In particular, it must always be guaranteed that machine shutdown can not be prevented or delayed once the command for shutdown has been issued. The engine shutdown command must always have priority over the start command.

If machines or machine components are designed to interact, they must be designed and built such that all devices can shut down. This relates to:

- Emergency-stop command devices for the machine itself (engine, engine-generator set)
- All connected devices insofar as their continued operation could result in a hazard

Important

Engines with exhaust aftertreatment: When the emergency stop is actuated on engine with an exhaust aftertreatment system, note that the emergency-stop function also interrupts the reducing agent supply. This can result in overheating of the reducing agent dosing units. For details and further information, see Operating Instructions.

1.5 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.